

Data Modeler – Power BI Data Modeling Project

Project Overview

This project is based on Power BI Data Modeling.

The main objective of this project was to create a proper relational data model using fact tables and dimension tables.

The project was completed using:

- Power Query Editor
- Model View
- Matrix Tables

What I Did In This Project

1. Imported Data

I imported multiple Excel files into Power BI:

- Sales_Fact
- Returns_Fact
- Customer_Dim
- Product_Dim
- Region_Dim
- Date_Dim

2. Data Cleaning Using Power Query

In Power Query Editor I performed:

- Removed blank rows
- Corrected data types
- Checked empty values
- Cleaned unnecessary data
- Loaded transformed data into the model

3. Created Relationships

I connected all tables using Primary Keys and Foreign Keys.

Relationships created:

- Customer_Dim → Sales_Fact
- Product_Dim → Sales_Fact
- Region_Dim → Sales_Fact
- Date_Dim → Sales_Fact

- Sales_Fact → Returns_Fact
- Date_Dim → Returns_Fact (Inactive Relationship)

4. Created Star Schema

I used Sales_Fact as the central fact table and connected all dimension tables around it:

- Customer_Dim
- Product_Dim
- Region_Dim
- Date_Dim

5. Implemented Snowflake Concept

Returns_Fact was added as an additional related table to demonstrate advanced relationship handling and inactive relationships.

6. Created Hierarchies

Date Hierarchy:

Year → Quarter → Month → Date

Region Hierarchy:

Country → State → City

Product Hierarchy:

Category → Subcategory → ProductName

Hierarchies help in drill-down analysis and better data organization.

7. Data Formatting

I applied proper formatting:

- Revenue → Currency
- Quantity → Whole Number
- Discount → Decimal
- Date → Date Format

8. Model Enhancement

- Hide foreign key columns from report view
- Arranged tables neatly
- Aligned model properly

9. Verification Using Matrix Tables

To verify relationship flow and filtering, I created matrix tables for:

- Sales by Product Category and Region
- Return Reasons by Fiscal Year
- Revenue by Customer Segment

Fact tables used:

- Sales_Fact
- Returns_Fact

Dimension tables used:

- Customer_Dim
- Product_Dim
- Region_Dim
- Date_Dim

Cardinality

Cardinality defines how tables are related to each other in a data model.

Types of Cardinality:

1 : 1 Relationship

One record in the first table is connected to only one record in the second table.

1 : Many Relationship

One record in the first table is connected to multiple records in the second table.

Many : 1 Relationship

Multiple records in the first table are connected to one record in the second table.

Many : Many Relationship

Multiple records in one table are connected to multiple records in another table.

Primary Keys and Foreign Keys

Primary Keys

- Customer_Dim → CustomerID
- Product_Dim → ProductID
- Region_Dim → RegionID
- Date_Dim → DateID

Foreign Keys

Sales_Fact:

- CustomerID → Customer_Dim.CustomerID
- ProductID → Product_Dim.ProductID

- RegionID → Region_Dim.RegionID
- DateID → Date_Dim.DateID

Returns_Fact:

- OrderID → Sales_Fact.OrderID
- DateID → Date_Dim.DateID (Inactive Relationship)

Star Schema

A Star Schema is a model where one central fact table connects directly to multiple dimension tables.

Snowflake Schema

Snowflake Schema is an extended version of star schema where additional related tables are connected.

Inactive Relationship

An inactive relationship exists in the model but is not actively used for filtering.

In this project:

Date_Dim → Returns_Fact was created as an inactive relationship.

Conclusion

This project successfully demonstrates Power BI Data Modeling concepts including relationship management, schema design, hierarchy creation, and data transformation using Power Query and Model View.